

Calculus with Polar Coordinates

MATH 146 – Calculus II

Torin Quinlivan

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Recall: Areas in Calculus

The area formula for *Cartesian functions* $y = f(x)$ depends on the area formula for rectangles and the limit of the Riemann sum.

Definition

Let $f(x)$ be continuous, nonnegative function on an interval $[a, b]$, and let $A_n \sum_{i=1}^n f(x_i^*) \Delta x$ be a Riemann sum for $f(x)$. Then the area under the curve $y = f(x)$ on $[a, b]$ is given by

$$A = \lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \int_a^b f(x) dx.$$

Polar Areas in Calculus

The area formula for *polar functions* $y = f(x)$ depends on the area formula for *circles* and the limit of the Riemann sum.

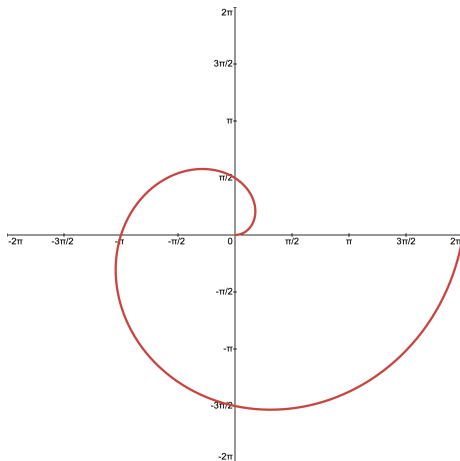
Definition

Let $f(\theta)$ be continuous, nonnegative function on an interval $[\alpha, \beta]$, and let $A_n = \sum_{i=1}^n \frac{1}{2}(\Delta\theta)(f(\theta_i^*))^2$ be a Riemann sum for $f(x)$. Then the area under the curve $r = f(\theta)$ on $[\alpha, \beta]$ is given by

$$A = \lim_{n \rightarrow \infty} A_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \frac{1}{2} \int_{\alpha}^{\beta} (f(\theta))^2 d\theta.$$

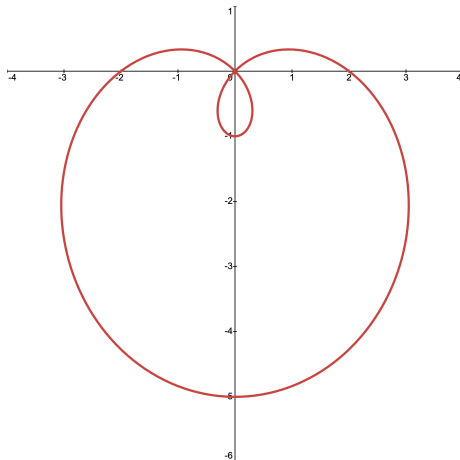
Example: Polar Areas

Example: Find the area within the polar function $f(\theta) = \theta$, where $0 \leq \theta \leq 2\pi$.



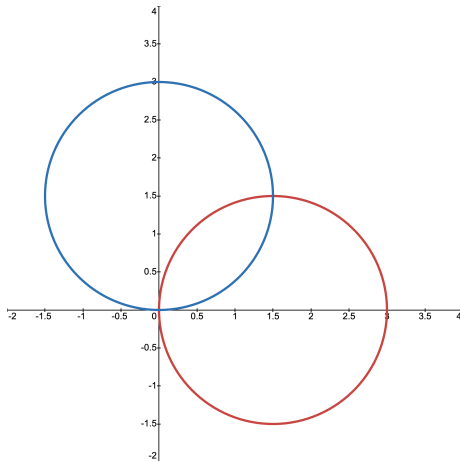
Example: Polar Areas

Example: Find the area enclosed by the inner loop of $f(\theta) = 2 - 3\sin(\theta)$.



Example: Polar Areas

Example: Find the region common to $f(\theta) = 3 \cos(\theta)$ and $g(\theta) = 3 \sin(\theta)$.



Polar Derivatives

If $r = f(\theta)$, recall that:

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

What is the slope of the tangent line $\frac{dy}{dx}$?

Arc Length in Parametric Coordinates

Theorem (Arc Length in Parametric Coordinates)

Given differentiable parametric functions $x(t)$ and $y(t)$, the arc length of the curve given parametric by x and y over the interval $[a, b]$ is

$$\int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt.$$

Example: Arc Length in Parametric Coordinates

Example: Find the circumference of a circle of radius $r = 4$.

Arc Length in Polar Coordinates

$$x'(\theta) = f'(\theta) \cos(\theta) - f(\theta) \sin(\theta)$$

$$\begin{aligned}(x'(\theta))^2 &= (f'(\theta) \cos(\theta) - f(\theta) \sin(\theta))^2 \\ &= (f'(\theta))^2 \cos^2(\theta) - 2f(\theta)f'(\theta) \cos(\theta) \sin(\theta) + (f(\theta))^2 \sin^2(\theta)\end{aligned}$$

and

$$y'(\theta) = f'(\theta) \sin(\theta) + f(\theta) \cos(\theta)$$

$$\begin{aligned}(y'(\theta))^2 &= (f'(\theta) \sin(\theta) + f(\theta) \cos(\theta))^2 \\ &= (f'(\theta))^2 \sin^2(\theta) + 2f(\theta)f'(\theta) \cos(\theta) \sin(\theta) + (f(\theta))^2 \cos^2(\theta)\end{aligned}$$

Arc Length in Polar Coordinates

Combining these:

$$\begin{aligned}(x'(t))^2 + (y'(t))^2 &= (f'(\theta))^2 \cos^2(\theta) - \cancel{2f(\theta)f'(\theta)\cos(\theta)\sin(\theta)} \\ &\quad + (f(\theta))^2 \sin^2(\theta) + (f'(\theta))^2 \sin^2(\theta) \\ &\quad + \cancel{2f(\theta)f'(\theta)\cos(\theta)\sin(\theta)} + (f(\theta))^2 \cos^2(\theta) \\ &= (f(\theta))^2 (\sin^2(\theta) + \cos^2(\theta)) \\ &\quad + (f'(\theta))^2 (\sin^2(\theta) + \cos^2(\theta)) \\ &= (f(\theta))^2 + (f'(\theta))^2\end{aligned}$$

Arc Length in Polar Coordinates

Theorem (Arc Length in Polar Coordinates)

Let f be a function whose derivative is continuous on an interval $\alpha \leq \theta \leq \beta$. The length of the graph of $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is

$$\begin{aligned} L &= \int_{\alpha}^{\beta} \sqrt{(x'(\theta))^2 + (y'(\theta))^2} dt \\ &= \int_{\alpha}^{\beta} \sqrt{(f(\theta))^2 + (f'(\theta))^2} dt \\ &= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} dt \end{aligned}$$